All six models use conventional OLS, exclude **Adult Mortality**, and include no country or year fixed effects. Overall models control for log GDP per capita, Schooling, and development status; within-group models control for the first two.

Let X denote government healthcare expenditure as a percentage of total government expenditure. In the table, b is the coefficient on X, and c is the coefficient on the additional nonlinear term.

Model	Healthcare expenditure specification	b	c	p-value for c	R^2	Observations
Overall quadratic	$bX + cX^2$	0.161948	0.001216	0.6141	0.695015	2,702
Overall linear plus logarithm	$bX + c\log(X)$	0.176937	0.149959	0.7621	0.694997	2,702
Overall linear plus reciprocal	$bX + c/X$	0.196036	0.216609	0.8768	0.694989	2,702
Overall piecewise linear	$bX + c\max(X - 8.7600, 0)$	0.270742	−0.112662	0.1372	0.695236	2,702
Developed-country quadratic	$bX + cX^2$	−0.219782	0.013506	**0.0129**	0.740654	496
Developing-country quadratic	$bX + cX^2$	0.151079	0.002536	0.3385	0.599440	2,206

None of the additional nonlinear terms in the overall models is significant at the 10% level. The developed-country quadratic term is significantly positive, indicating increasing rather than diminishing slopes. The developing-country quadratic term is not significant. Therefore, these results do **not consistently support diminishing marginal returns**.

For the piecewise model, the estimated slopes are:

Quantity	Estimate	95% confidence interval	p-value
Slope below 8.76%	0.270742	[0.157317, 0.384166]	<0.001
Slope above 8.76%	0.158080	[0.092073, 0.224087]	<0.001
High-expenditure slope minus low-expenditure slope	−0.112662	[−0.261232, 0.035908]	0.1372

Both slopes are significantly positive, but their difference is not statistically significant.

For the next stage, we will use the **piecewise linear specification with lagged healthcare expenditure**:

$$\begin{aligned} LifeExpectancy_{it} = \beta_0 + \beta_1 X_{i,t-L} + \beta_2 \max(X_{i,t-L} - k, 0) \\ + \gamma_1 \log(GDPpc_{it}) + \gamma_2 Schooling_{it} + \gamma_3 Developed_i + u_{it}, \end{aligned}$$

where L is the lag length.

We choose this specification because:

- **It is easy to interpret:** β_1 measures the low-expenditure slope, while $\beta_1 + \beta_2$ measures the high-expenditure slope.
- **It directly tests slope attenuation:** a negative β_2 indicates a smaller slope at higher expenditure levels, without imposing that result.
- **It has the best in-sample fit among the four overall nonlinear models**, with the lowest AIC and BIC. However, its current curvature evidence remains insufficient.

We will compare contemporaneous, one-year-lagged, and three-year-lagged expenditure using a **common sample and a fixed threshold**. This will help assess whether the association emerges with a delay. Lagging expenditure alone does not establish causality, and all lag results should be reported rather than selecting only significant ones.

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